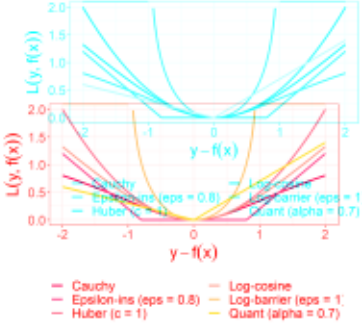
Advanced Risk Minimization

Advanced Regression Losses



Learning goals

- Know the Huber loss
- Know the log-cosh loss
- Know the Cauchy loss
- Know the log-barrier loss
- Know the ϵ-insensitive loss
- Know the quantile loss

ADVANCED LOSS FUNCTIONS

Special loss functions can be used to estimate non-standard posterior components, to measure errors in a custom way or are designed to have special properties like robustness.

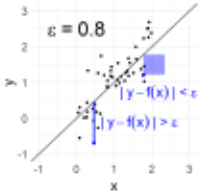
Examples:

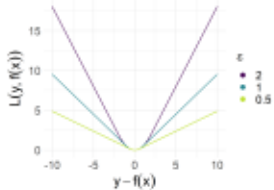
- Quantile loss: Overestimating a clinical parameter might not be as bad as underestimating it.
- Log-barrier loss: Extremely under- or overestimating demand in production would put company profit at risk.
- ϵ-insensitive loss: A certain amount of deviation in production does no harm, larger deviations do.

HUBER LOSS

$$L(y,f) = \begin{cases} \frac{1}{2}(y-f)^2 & \text{if } |y-f| \leq \epsilon \\ \epsilon|y-f| - \frac{1}{2}\epsilon^2 & \text{otherwise} \end{cases}, \quad \epsilon > 0$$

- Piece-wise combination of $L1/L2$ to have robustness/smoothness
- Analytic properties: convex, differentiable (once)





- Risk minimizer and optimal constant do not have a closed-form solution. To fit a model numerical optimization is necessary.
- Solution behaves like **trimmed mean**: a (conditional) mean of two (conditional) quantiles.

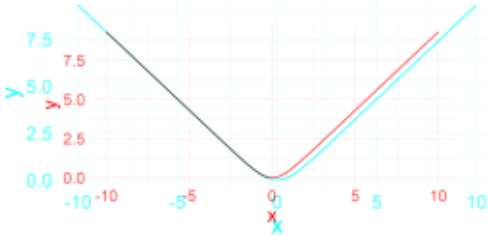
LOG-COSH LOSS

R. A. Saleh and A. Saleh 2022

$$L(y, f) = \log(\cosh(|y - f|)) \quad \text{where} \cosh(x) := \frac{e^x + e^{-x}}{2}$$

- Logarithm of the hyperbolic cosine of the residual.
- Approximately equal to $0.5(|y - f|)^2$ for small f and to $|y - f| - \log 2$ for large f, meaning it works mostly like $L2$ loss but is less outlier-sensitive.
- Has all the advantages of Huber loss and is, moreover, twice differentiable everywhere.

- Logarithm of the hyperbolic cosine of the residual.
- Approximately equal to $0.5(|y - f|)^2$ for small residuals and to $|y - f| - \log 2$ for large residuals, meaning it works a smoothed out $L1$ loss using $L2$ around the origin.
- Has all the advantages of Huber loss and is, moreover, twice differentiable everywhere.



LOG-COSH LOSS ▸ R. A. Saleh and A. Saleh 2022 / 2

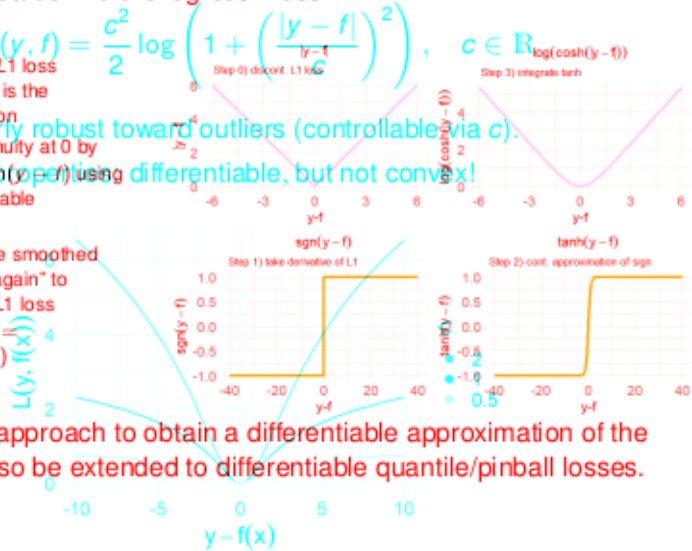
What is the idea behind the log-cosh loss?

Essentially, we

1. take derivative of $L1$ loss w.r.t. $y - f$, which is the $\text{sign}(y - f)$ function
2. eliminate discontinuity at 0 by approximating $\text{sign}(y - f)$ using the cont. differentiable $\tanh(y - f)$
3. finally integrate the smoothed sign function "up again" to obtain smoothed $L1$ loss $\log(\cosh(y - f)) = \log(\cosh(|y - f|))$

$$L(y, f) = \frac{c^2}{2} \log\left(1 + \left(\frac{|y - f|}{c}\right)^2\right), \quad c \in \mathbb{R}_{\log(\cosh(y-f))}$$

- Particularly robust toward outliers (controllable via c).
- Analytically optimizing differentiable, but not convex!



The log-cosh approach to obtain a differentiable approximation of the $L1$ loss can also be extended to differentiable quantile/pinball losses.

The cosh(θ, σ) distribution:

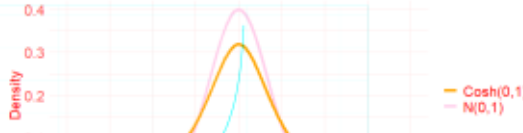
The (normalized) reciprocal cosh(x) defines a pdf by its positivity on $\mathbb{R}$ and since $\int_{-\infty}^{\infty} \frac{1}{\pi \cosh(x)} dx = 1$.

We can define a location-scale family of distributions (using θ and σ) resembling Gaussians with **heavier tails**.

It is easy to check that ERM using the log-cosh loss is equivalent to MLE of the cosh$(\theta, 1)$ distribution.

- $p(x|\theta, \sigma) = \frac{1}{\pi\sigma \cosh\left(\frac{x-\theta}{\sigma}\right)}$
- $\mathbb{E}_{X\sim p}[X] = \theta$
- $\mathrm{Var}_{X\sim p}[X] = \frac{1}{4}(\pi^2\sigma^2)$
- $\hat{\theta}^{MLE} = \arg\max_\theta \prod_{i=1}^{n} \frac{1}{\pi \cosh(x_i - \theta)} \equiv$

  $\hat{\theta} = \arg\min_\theta \sum_{i=1}^{n} \log(\cosh(x_i - \theta))$

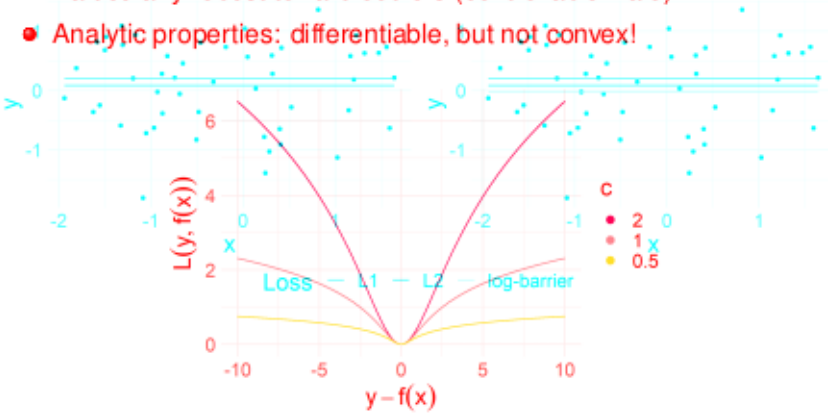


CAUCHY LOSS

- Note that the optimization problem has no (finite) solution if there is no way to fit a constant where all residuals are smaller than ϵ.

$$L(y, f) = \frac{c^2}{2} \log\left(1 + \left(\frac{|y - f|}{c}\right)^2\right), \quad c \in \mathbb{R}$$

- Not feasible for $\epsilon = 1$ Feasible for $\epsilon = 2$
- Particularly robust toward outliers (controllable via c).
- Analytic properties: differentiable, but not convex!



QUANTILE LOSS / PINBALL LOSS

$$L(y,f) = \begin{cases} (1-\alpha)(f-y) & \text{if } y < f \\ \alpha(y-f) & \text{if } y \geq f \end{cases}, \quad \alpha \in (0,1)$$

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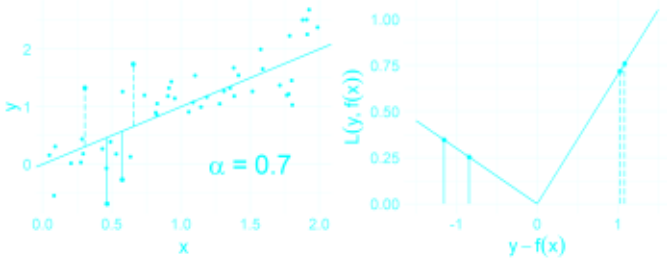
- Extension of $L1$ loss (equal to $L1$ for $\alpha = 0.5$).
- Weights either positive or negative residuals more strongly.
- $\alpha < 0.5$ ($\alpha > 0.5$) penalty to over-estimation (under-estimation)
- Risk minimizer is (conditional) α-quantile (median for $\alpha = 0.5$).